



The autoignition of iso-dodecane in low to high temperature range: An experimental and modeling study

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ABSTRACT

2,2,4,6,6-pentamethylheptane, a highly branched alkane, is a promising component candidate of branched alkanes in the surrogates for jet fuels and a major component in alternative fuels. In spite of its great relevance and importance in real fuels, it has only attracted very little interest. This work provides a new set of experimental data and a newly proposed detailed kinetic mechanism for this hydrocarbon. In the experiments, the autoignition characteristics of 2,2,4,6,6-pentamethylheptane, was measured in a rapid compression machine and a shock tube spanning over the equivalence ratios of 0.5, 1 and 1.5, pressures of 15 and 20 bar and temperature range of 603–1376 K. The dependence of ignition delay times on pressure, mole fraction of fuel, mole fraction of oxygen and dilution ratio were systematically investigated. Negative temperature coefficient (NTC) behavior was observed during the autoignition event, but the NTC temperature window was found to be much lower than the counterpart normal paraffin. A detailed kinetic model containing 729 species and 3390 reactions was proposed to describe the combustion behavior of this compound over low-to-high temperature range. The model was validated against the present experimental data, as well as the limited datasets in literature. Good agreements were observed between the experimental data and the predictions from the proposed model. Kinetic analyses, including sensitivity analysis and reaction pathway analysis, were carried out to provide insight into the combustion characteristics from the kinetic perspective. Comparisons were also carried out with the datasets for other branched alkanes and normal-paraffin with same carbon numbers over the same conditions to reveal the effects of the molecular structure.

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1. Introduction

Comprehensively validated kinetic models of fuels potentially promise the accurate simulation of the combustion in engines and provide insight into the effect of fuel composition on the performance of engines. Therefore, with the increasingly stringent demand for efficiency and emissions, reliable kinetic models play a growingly significant role in the design of high-efficiency and low-emission engines and the development of advanced combustion technologies. Given the complex composition of real fuels (e.g., jet fuels), mixtures of limited well-studied components, usually called surrogate fuels, are utilized to describe the combustion behavior of these fuels. Branched alkanes account for around 35–40% (wt.%) of all hydrocarbons present in conventional jet fuels [1–3] and even higher quantities in non-petroleum-derived alternative fuels [4–8], and thereby have a great influence on the reactivity of these fuels.

Consequently, it is essential to include representative branched alkanes in the surrogate palette.

In the previous studies, iso-octane or iso-cetane or the combination of the two have been extensively used in jet fuels surrogate formulation. However, the carbon numbers of the two compounds are outside the range of 10–14 [3,9], the average carbon number of isoparaffin in jet fuels. Hence, this would restrict the emulation of properties that depend on the molecular size, e.g., molecular weight, distillation curve and diffusion flame extinction [9,10]. This situation is true in the surrogate fuels proposed by Kim et al. [11,12]. The addition of an isoalkane with molecular weight and boiling range similar to the average of target fuel to the surrogate palette would be beneficial for the better match. Iso-dodecane (viz 2,2,4,6,6-pentamethylheptane, PMH) is a highly branched alkane of low reactivity and satisfies the need of similar molecular weight to jet fuels, and thus it would be a good alternative to iso-octane and iso-cetane. There have been some studies using PMH as a surrogate component [6,9,13–15]. In the study of Kim et al. [9], they found that PMH is an attractive surrogate candidate component for a kind of coal-derived jet fuel and recommended it as the next

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generation jet fuel surrogate component candidate. In addition to a potential surrogate component for jet fuels, PMH is also one of the major components of the promising alternative fuels, e.g., Gevo alcohol to jet fuel [5,8,14,15], hydrotreated renewable jet fuel [13], hydrotreated Vegetable Oil [6]. Therefore, it is significantly important to investigate the combustion kinetics of PMH.

Although PMH could be a good choice for representing the branched alkane class, compared with the extensively studied reference fuels iso-octane and iso-cetane, only very little attention was paid to PMH, especially on the combustion behavior. There are only a few studies that explored the combustion behavior of PMH. Among them, Min et al. [8] measured the ignition delay times (IDTs) of PMH over equivalence ratios of 1 and 0.5, pressures of 10 and 20 bar using the rapid compression machine (RCM) at UIUC. They compared the IDTs of PMH with those of n-dodecane and 2,9-dimethyldecane, and found that PMH showed much lower reactivity. Won et al. [10] evaluated the combustion characteristics of PMH by comparing its IDTs and strained diffusion flame extinction with those of iso-octane, iso-cetane and 50/50% molar blend of iso-octane/iso-cetane. They concluded that the combustion behavior of PMH can be closely emulated by the iso-octane/iso-cetane blends. Other studies included the measurements of physical properties of PMH and its application as a surrogate component [13–15], as well as the performance of binary fuel blends of PMH included in engines [6]. To our knowledge, there are no available kinetic models for PMH. Naik et al. [16] claimed that they proposed detailed mechanisms for over 60 compounds, including PMH. However, open-access is not provided for the library of these kinetic models. Since one of the criteria to determine the surrogate palette is the availability of mature chemical mechanisms, the absence of kinetic mechanism will limit its application as a surrogate component. In summary, the studies related to PMH are far from sufficient, including both experimental measurements and the development of kinetic models.

It was well established that new experimental data is an important step toward developing a reliable kinetic model. Aiming at the insufficiency of studies on PMH, the objective of the present work is to provide a newly measured experimental dataset and to develop a detail kinetic model for PMH. We measured the autoignition characteristics of PMH over low-to-high temperature range by combining the RCM and shock tube (ST) at Shanghai Jiao Tong University (SJTU). The newly measured IDTs span over the equivalence ratios of 0.5, 1 and 1.5, and pressures of 15 and 20 bar over a temperature range of 603–1376 K. A kinetic mechanism including 729 species and 3390 reactions was proposed and validated against the present experimental data, as well as the limited dataset in the literature. Kinetic analyses, including brute force sensitivity analysis and important reaction pathways, were performed to provide insight into the combustion chemistry of PMH.

2. Methods

2.1. RCM measurement

Since the details of the heated RCM at SJTU, including the structure of the setup and the procedure of operation, can be found elsewhere [1,17–24], it will not be repeated here. The temperature of the end of compression (EOC) was determined according to the adiabatic core hypothesis [25]. In the RCM measurements, a minimum of 5 parallel runs was performed for each initial condition to ensure the repeatability. In addition, a non-reactive run was conducted closely following the corresponding reactive run by replacing O_2 with equivalent N_2 to emulate the heat loss of the facility. Figure 1 shows a typical pressure trace in the RCM measurement, including the reactive and non-reactive one. It can be seen that the target fuel shows an obvious two-stage ignition, namely the

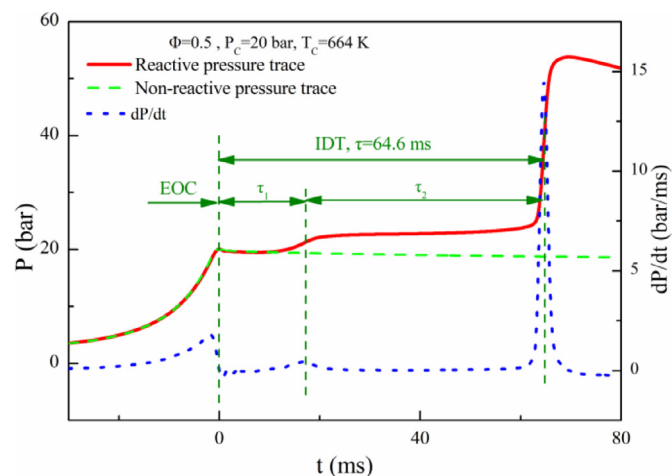


Fig. 1. Definition of IDT in RCM measurement.

first-stage (τ_1) and second-stage (τ_2) ignition. The non-reactive pressure trace closely overlaps with the reactive one before the first-stage ignition, implying no notable exothermicity during the compression stroke. Hence, it is acceptable to emulate the heat loss during the compression stroke using the non-reactive run. In order to be consistent with the previous studies, the IDTs of RCM experiments were defined as the period from EOC to the maximum pressure rise rate, as shown in Fig. 1. Since the apparatus and experimental method is the same as [1,17–19], the systematic uncertainty is estimated to be within $\pm 10\%$ as well.

2.2. ST measurement

The heated ST at SJTU had been described in our previous publication [1,17,21,26–28]. Therefore, here we only present a brief introduction. The inner diameter of the ST is 90 mm, and the lengths of the driver section and driven section are 4 m and 5 m, respectively. For the tests with IDT shorter than 2 ms, pure He was used as driver gas. PMH was expected to be a fuel of low reactivity according to its low cetane number, thus tailored He/ N_2 mixtures were used as driver gas to extend the effective experimental time. Before each run, the composition of driver gas was determined based on the target initial temperature and pressure. Throughout the experiments, the intersection of the baseline and the tangent at the maximum slope of the profile of the OH^* signal recorded by a photomultiplier is taken as the onset of the ignition. The IDT in ST measurements was defined as the time from the arrival of the incident shock wave at the endwall to the onset of ignition, as shown in Fig. 2. The experimental uncertainty is estimated to be $\pm 20\%$ based on Arrhenius-type correlation using the root-sum-squares method [29,30].

2.3. Mixture preparation and test conditions

The sample used in the present study is provided by TCI with a purity of $>98.0\%$ (GC). Since PMH is a large hydrocarbon of high boiling point, a preheating temperature of $120^\circ C$ (for stoichiometric mixtures in RCM measurement, $110^\circ C$ was used) was used to ensure the complete vaporization of the fuel. The monitoring results showed that the vaporization rate was higher than 99% throughout the experiments. Before the first experimental run, the prepared mixture was maintained at the preset temperature for at least 3 h to ensure homogeneity. As shown in Table 1, the PMH/air mixtures with equivalence ratios of 0.5, 1.0 and 1.5 were measured in this work. In addition, different compositions of oxidizer were adopted to investigate the effect of dilution ratios.

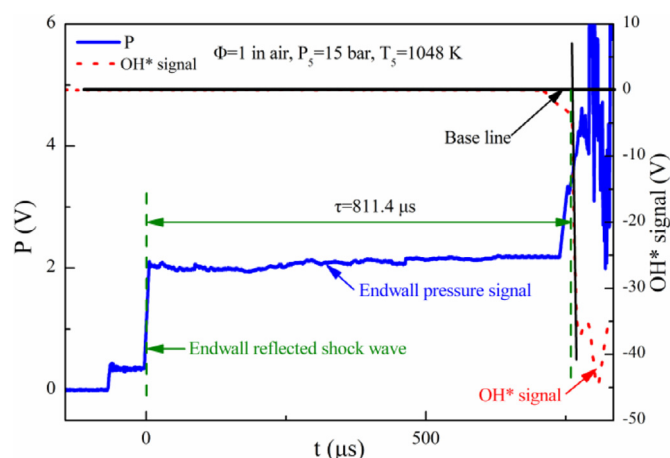


Fig. 2. Definition of IDT in ST measurement.

Table 1

Test conditions used in the RCM measurements for PMH.

Equivalent ratio	x_{fuel} (mol.%)	x_{O_2} (mol.%)	x_{N_2} (mol.%)	Dilution ratio ($x_{\text{N}_2}/(x_{\text{N}_2} + x_{\text{O}_2})$)	Pressure (bar)
0.5	0.56	20.88	78.55	79%	15/20
1.0	1.12	20.76	78.11	79%	15/20
1.5	1.67	20.65	77.68	79%	15/20
1.0	1.67	30.97	67.35	68.5%	15/20

2.4. Numerical methods

The experimental data was simulated using the closed homogeneous batch reactor model in Chemkin-pro software with the proposed kinetic model in this study. For both RCM and ST measurements, constant Volume (CONV) simulation was used to predict the IDTs. However, in order to mimic the facility effect, additional RCM simulations using the volume histories derived from the non-reactive runs were performed for RCM measurements. For ST measurements, an average pressure rise rate of 3% /ms was taken into account by using the Senkin/VITM approach [31].

3. Kinetic model development

To our knowledge, there is no published detailed mechanism for PMH to date. In the present study, we proposed a mechanism including both low temperature and high temperature reaction scheme. The AramcoMech 2.0 [32] was chosen as the C0-C4 core mechanism, while the sub-mechanism of this fuel was developed in the same approach as that employed by previous researches for large normal-alkanes and branched alkanes [33–35], namely using the 30 reaction classes proposed by Sarathy et al. [36]. Since the kinetics of PMH has seldom been investigated, the rate coefficients of reactions in the present model were mainly estimated based on analogies or assigned according to the rate rules, or taken from experimental measurement or theoretical calculation if available.

Figure 3 compares the molecular structures of PMH with two other branched alkanes. It can be found that there is a high simi-

larity among the molecular structures of PMH, iso-octane and iso-cetane, therefore, it is reasonable to speculate that there will be similar radical pool formation during their combustion process and thereby a similar reactivity between these fuels. This is true in the recent study by Kim et al. [9], who reported that the combustion characteristics of PMH is very similar to the latter two. Therefore, the development of the sub-mechanism of PMH mainly referred to the kinetic mechanisms of these two hydrocarbons [35,37]. The final kinetic model includes 729 species and 3390 reactions. The nomenclature glossary of the important intermediate species is provided in the Supplemental Material. The thermodynamic data of the species mostly adopted from iso-cetane or iso-octane mechanisms [35,37], while the newly presented species in this mechanism were estimated using the group additivity method. Further details about the reaction mechanism will be introduced in the following sections.

3.1. High temperature reaction pathway

As shown in Fig. 3, PMH is a highly branched alkane with five methyl side branches, which increases the complexity of the molecular structure. However, the entirely mirror-symmetric structure, makes the reaction scheme briefer than expected. For instance, there are only five but not twelve distinct carbon sites as shown in Fig. 3, where H atoms connected to can be abstracted, leading to five fuel-derived radicals.

Unimolecular decomposition and H atom abstraction from fuel by active radicals, as well as the beta-scission of the fuel-derived radicals are significant reaction classes in the high temperature regime and play significant roles in the combustion chemistry. For the unimolecular decomposition, both the dissociation of C–C bonds and C–H bonds were included. In this work, the rate coefficients of the unimolecular decomposition and decomposition of fuel derived radicals were estimated based on the analogies in the isocetane and iso-octane mechanism [35,37]. For H abstraction reactions, ten abstractors were considered in the present mechanism. Rate coefficients of this reaction class (except H abstraction by HO_2 radical) were estimated according to the analogies in the iso-cetane model [36]. H abstraction by HO_2 contributes little to the oxidation of fuel in low temperature range, but this reaction type becomes very important at higher temperatures, especially in the intermediate temperature range, where the decomposition of H_2O_2 dominates the overall reactivity. We prescribed the rate coefficients for H abstraction from parent fuel by HO_2 according to the rate rules proposed by Iparraguirre et al. [38].

3.2. Low temperature reaction pathway

The chain branching reaction sequences of low temperature start from the O_2 addition to fuel derived radicals to produce alkylperoxy radicals (ROO), and then is followed by reaction classes in the order of isomerization to hydroperoxy-alkyl radicals (QOOH), second O_2 addition to form peroxy alkylhydroperoxide (OOQOOH) and the isomerization of OOQOOH to ketohydroperoxides, and ultimately is finalized by the decomposition of ketohydroperoxides. In our model, the rate coefficients for O_2 addition to fuel derived radicals were assigned according to the rate rules proposed by Miyoshi

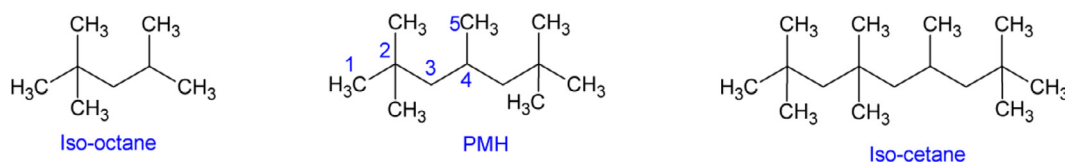


Fig. 3. Molecular structure of three branched alkane homologs. Numbers in PMH denote the serial numbers of carbons in this work.

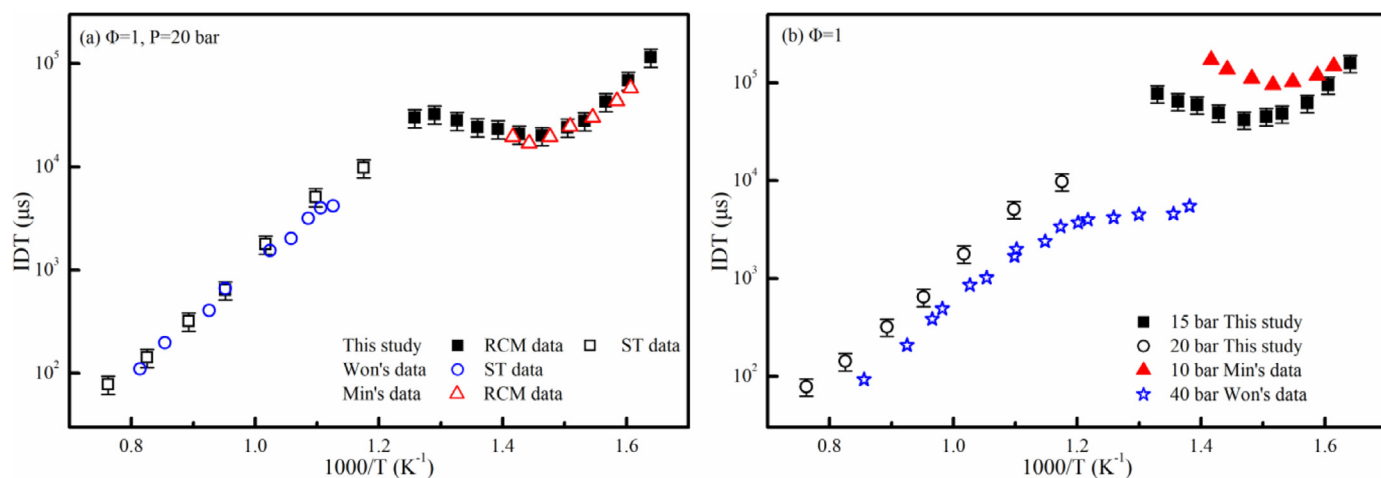


Fig. 4. Comparison with previous measured IDTs for PMH [8,10].

[39] based on carbon type. Isomerization of ROO via 4–8 membered ring transition states were included. The rate constants were taken from the analogies in the iso-cetane mechanism, but the pre-exponential factors were adjusted according to the degeneracy of hydrogen atoms. The rate constants for the second O_2 addition was prescribed to be one quarter of the first O_2 addition according to the suggestion of Bissoonauth et al. [40]. The rate constants for isomerization of OOQOOH to ketohydroperoxides were taken from analogies from the iso-cetane model, while for the subsequent decomposition of ketohydroperoxides, an activation energy of 39 kcal/mol was used.

In the low temperature regime, the most important competitive reaction classes include the concerted elimination of ROO, the formation of cyclic ether and beta-scission of QOOH. These reactions reduce the overall reactivity, as they produce stable alkenes and/or less-reactive radicals (e.g., HO_2) at the cost of consuming reactive radicals (usually OH, considering the contribution of H abstraction by OH radical at low temperatures). These reactions compete more effectively with the degenerated chain branching reactions as the temperature increases, and are generally believed to be responsible for the NTC behavior at intermediate temperatures. For concerted elimination reaction class, the rate rules suggested by Villano et al. [41] were used for reactions involving peroxy radicals connected to primary and secondary carbons, while the rate rules of Snitsiriwat and Bozzelli [42] were used for peroxy radical sited in the tertiary carbon. While for the cyclic ether formation and the decomposition of QOOH as well as the reactions of cyclic ethers with reactive radicals (OH and HO_2), the rate constants were referred to multiple literature [36,37].

Recent studies [35,43–45] found that the alternative isomerization of OOQOOH species to form $P(OOH)_2$ and the subsequent third O_2 addition reaction channel was important for the low temperature combustion and recommended the inclusion of this pathway in mechanisms of heavy hydrocarbons with more than 6 carbons. Therefore, the alternative isomerization and the subsequent reactions were included in our mechanism. For this reaction scheme, rate constants were prescribed based on analogies and estimations since very few studies focused on their kinetics. Specifically, for the alternative isomerization of OOQOOH, rate coefficients were assigned using the same rate rules as the isomerization of ROO. Three reaction pathways were proposed for the formed $P(OOH)_2$ species, namely, the beta-scission $P(OOH)_2$ =olefinic hydroperoxide+ HO_2 , the formation of hydroperoxy cyclic ether through $P(OOH)_2$ =hydroperoxy cyclic ether+OH, and the third O_2 addition to form OOP(OOH) $_2$. These reactions adopted the same rate rules as reaction classes $QOOH$ =alkene+ HO_2 , $QOOH$ =cyclic

ether+OH and the second O_2 addition, respectively. Subsequently, the OOP(OOH) $_2$ species is isomerized to keto-dihydroperoxide, and then keto-dihydroperoxide decomposed into small species. The rate coefficients for these reactions was assigned based on analogies of the formation and decomposition of ketohydroperoxides.

4. Results and discussion

4.1. Comparison with literature data

To date, studies on the autoignition of PMH are meager both at low and high temperature ranges. To evaluate the reliability of the experimental data, we specially measured the IDTs of the target fuel at the same operating condition as the previous studies. Figure 4 presents a comparison of the present experimental data with the datasets in literature [8,9] under the same or similar conditions. Figure 4(a) compares the IDTs measured at identical conditions. It can be seen that both present low and high temperature IDTs are in good agreement with the previous data. Figure 4(b) shows a comparison at different pressures, and the experimental data shows the same variation trend and the pressure dependence of IDTs is as expected.

4.2. Experimental results and model simulation

In the present study, a series of test conditions were chosen to cover a wide range of equivalence ratio, pressures, temperature and dilution ratios. In this section, we will systematically present the effect of the operating parameters on the autoignition characteristics of PMH and validate the proposed kinetic model against the newly obtained experimental data in this study.

Figure 5 shows a comparison of the IDTs of PMH at varied pressures. In whole, the plotted Arrhenius profiles can be divided into three regimes based on the slope, i.e., the low temperature regime, intermediate temperature regime and the high temperature regime. Typical NTC behavior can be observed in the intermediate temperature regime for this fuel under all the test conditions. It can also be seen that the IDTs decrease with the increasing pressure over the entire temperature range, and the pressure shows the most prominent effect in the NTC region. Such a phenomenon was extensively observed in the autoignition of hydrocarbons [1,17,24,46,47]. It is well established that the pressure dependent reaction $H_2O_2(+M)=OH+OH(+M)$ dominates the combustion chemistry in the intermediate temperature regime. Therefore, the rate coefficient of this reaction increases rapidly as pressure increases and compete less effectively with its competitive reaction

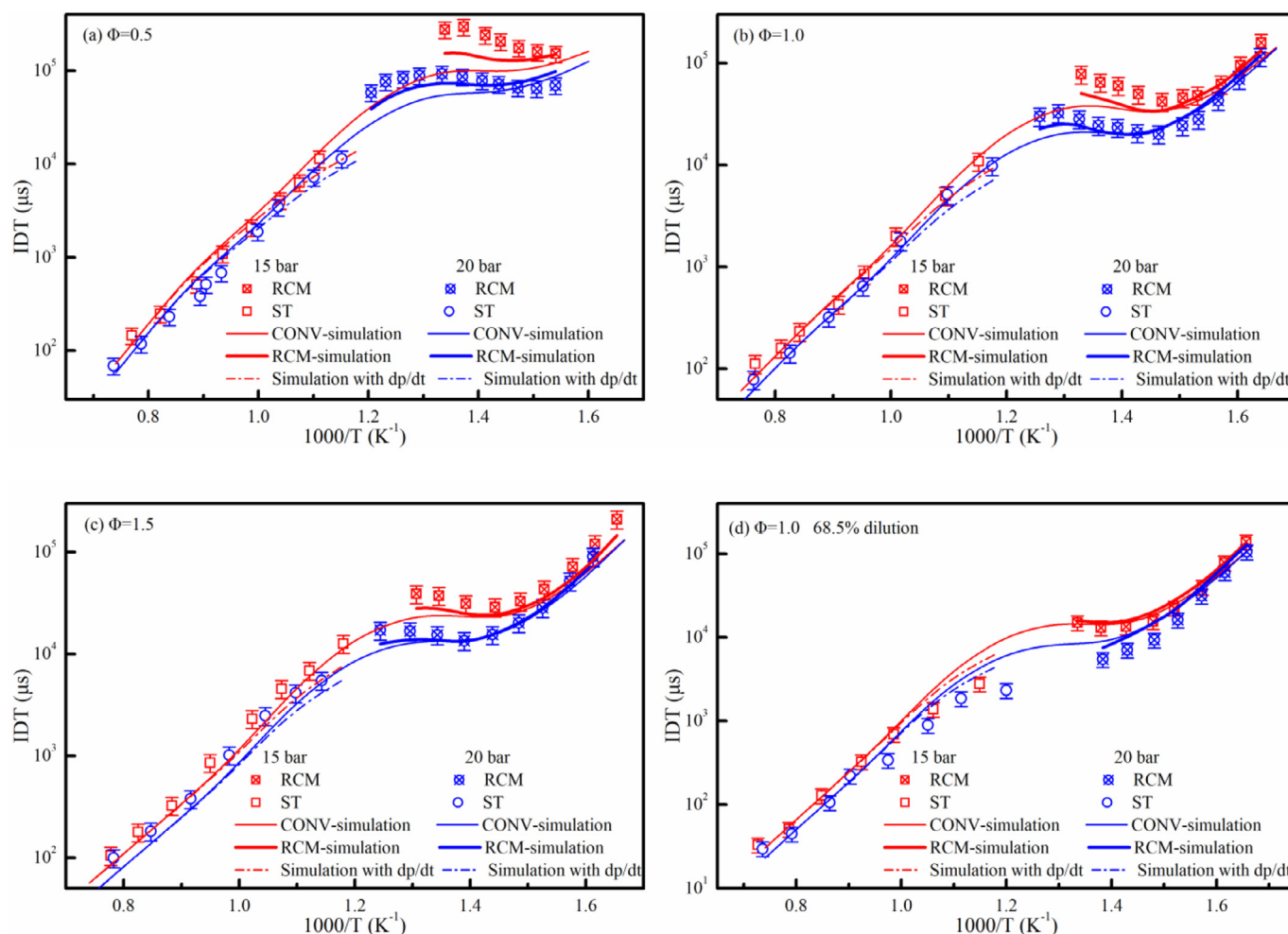


Fig. 5. Comparison of IDTs at different pressures.

pathways, enhancing the pressure effect. Consequently, IDT shows more pronounced dependence on the pressure in the NTC region compared to the low and high temperature region. A notable observation is a discrepancy by a factor of 2–3 between the RCM data and ST data in the NTC region, judging from the trend shown in Fig. 5(a) and (d). A similar phenomenon was reported in other fuels of low reactivity [35], but the reason is not clear now. Atef et al. [35] speculated that the invalidity of homogeneous core assumption in the determination of the temperature at EOC for RCM measurements and the inhomogeneity and boundary layer phenomenon in the shock tube may contribute to the difference. Future studies are of course needed to further reveal the mechanism for this discrepancy, but in-depth exploration is beyond the scope of this study. Figure 5 also shows the prediction from the proposed model, including the CONV and RCM simulations. The model follows the effect of pressure very well under all conditions, as well as the NTC behavior. For most test conditions, the predicted IDTs falls in the range of the experimental uncertainty. However, there are still some discrepancies between the simulations and experimental data. For instance, as shown in Fig. 5(a), the model underpredicts the IDTs of fuel-lean mixtures in low-to-intermediate temperature range under 15 bar; while for the 68.5% diluted mixtures, the model slightly overestimated the IDTs in low-to-intermediate temperature range in Fig. 5(d). Since the kinetics of PMH are seldom investigated, several rate constants used in the present model were estimated based on analogies and the model would be better to be treated as a priori. Therefore, such deviation is inevitable and in-depth kinetic investigation is needed

to improve the performance of the model. Overall, the model satisfactorily captures the autoignition characteristics of PMH.

The effect of equivalence ratio on IDT is shown in Fig. 6. When using air as the oxidizer, the equivalence ratio is varied by altering the fuel intake and oxygen mole fraction remains almost constant (around 20.75%). Hence, the effect of equivalence ratio here equals to the effect of fuel mole fraction. As the fuel mole fraction increases, the overall reactivity increases. Again, the effect of equivalence ratio is most notable in the NTC region. This is because the significance of the chain branching reaction ($\text{H}_2\text{O}_2(+\text{M})=\text{OH}+\text{OH}(+\text{M})$), which is subsequent to the H atom abstractions from the parent fuel by HO_2 , increases in NTC region. The kinetic model fairly follows the variation tendency of IDT with equivalence ratio over the entire temperature range. Another noticeable phenomenon is the distinct effects of fuel concentration for fuel-lean mixtures compared with fuel-rich mixtures, which is well predicted by the present model.

Figure 7 shows the influence of oxygen mole fraction on IDT. Over the entire temperature range, the reactivity increases with the increase of oxygen mole fraction. In low-to-intermediate temperature regime, the addition of oxygen to fuel-derived radicals, along with the subsequent degenerated chain branching reaction sequences control the combustion chemistry. Therefore, when the fuel remains constant, the decrease in oxygen concentration tends to prolong the induction period. While at high temperatures, the reaction $\text{H} + \text{O}_2 = \text{OH} + \text{O}$ governs the systematic reactivity, and thereby the IDT shows strong sensitivity to the variation of oxygen mole fraction. The present model well predicts the influence of

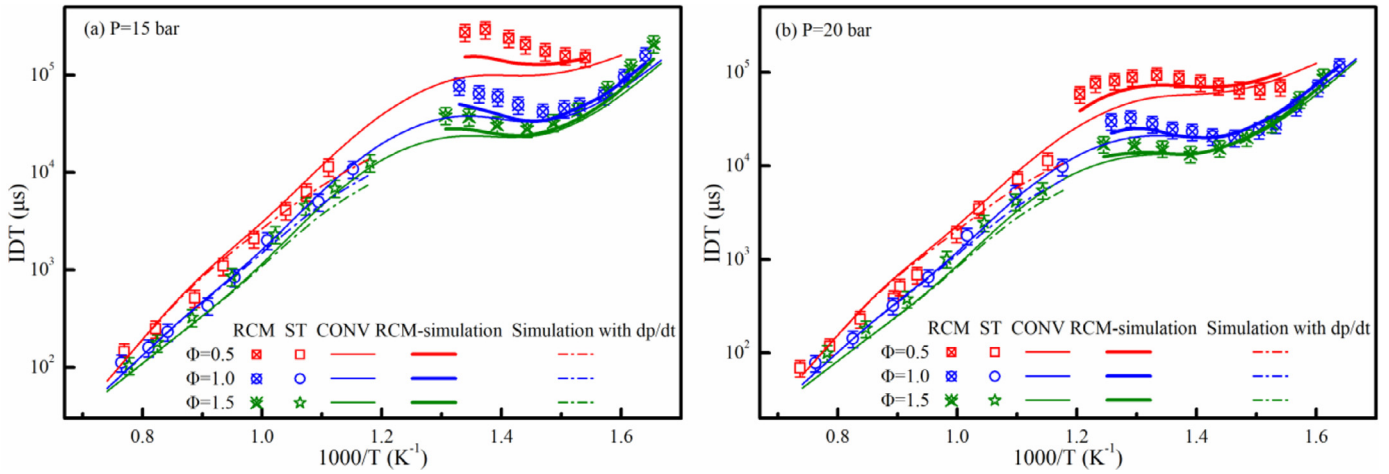


Fig. 6. Effect of equivalence ratio on IDT.

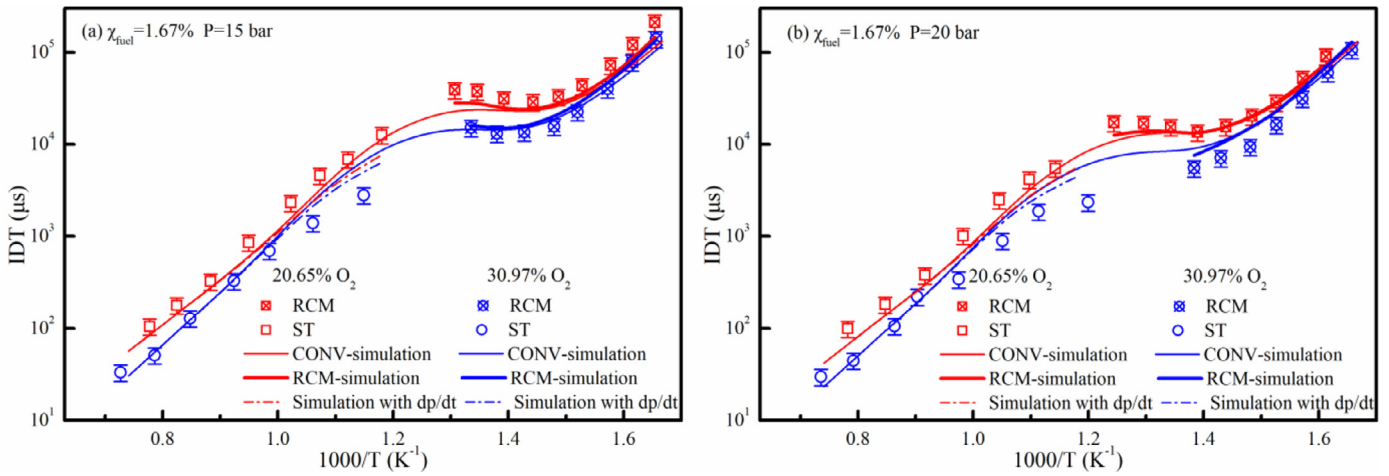


Fig. 7. Comparison of IDTs for mixture with the same fuel concentration but different oxygen concentration.

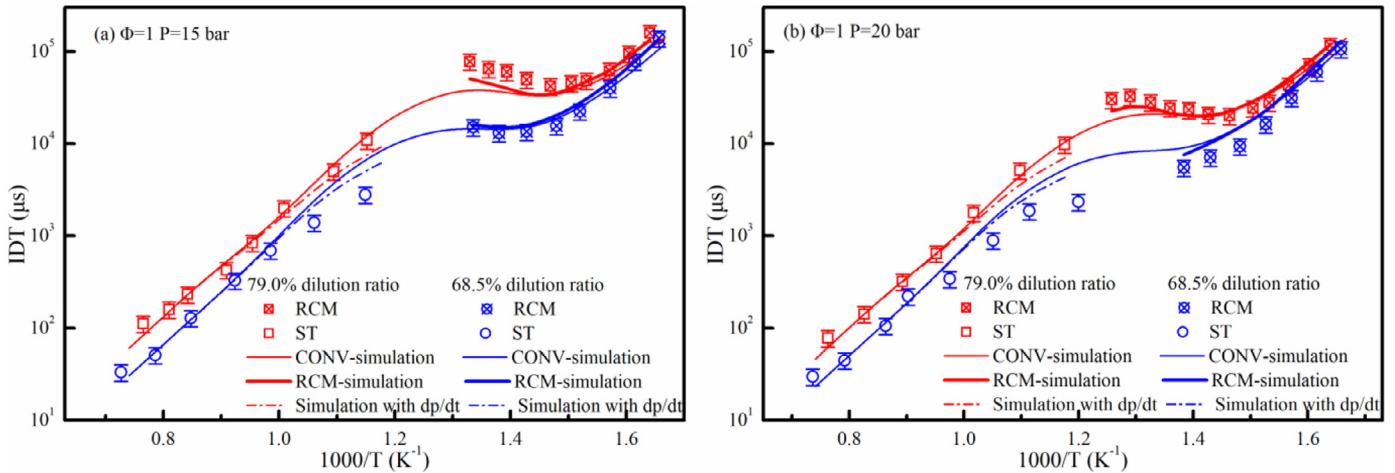


Fig. 8. Comparison of IDTs for stoichiometric mixtures with different dilution ratios.

oxygen in low-to-intermediate temperature region; however, the simulations show much weaker dependence on oxygen concentration in high temperature ranges compared to the experimental data.

Figure 8 compares the IDTs of stoichiometric PMH/O₂/N₂ mixtures with different dilution ratios. It is found that the increase of dilution ratio extends the IDTs notably, and the dilution ratio has the strongest effect on IDTs among the above-mentioned operating

parameters. The effect of dilution ratio is considered to be related to the variation of reactant partial pressure, as well as the thermodynamic properties, e.g., the specific heat capacity of the mixtures. Generally, the model captures the effect of the dilution ratio at an acceptable level.

Figure 9 further compares the experimental pressure traces with the simulated ones using RCM simulations at representative conditions. Under the two shown conditions, the current model

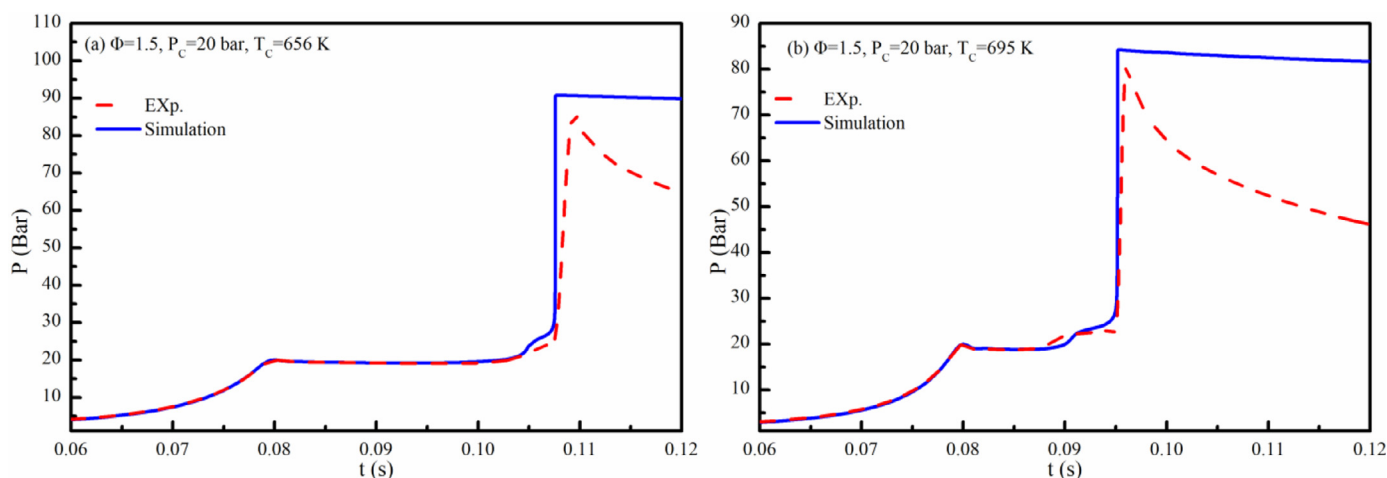


Fig. 9. Comparison of the simulated and experimental pressure trace for PMH in RCM measurements.

follows the experimental pressure history quite well before the ignition events. For instance, as shown in Fig. 9(a), the model well predicts both the first- and second-stage ignition delays. In addition, the pressure rise after the first-stage ignition is also well captured by the model. After the ignition event, due to the great discrepancy between the reactive and non-reactive runs, the effective volume derived from the non-reactive pressure trace fails to account for the heat loss. Consequently, the experimental traces show a much more remarkable pressure drop than the simulated ones.

Overall, the agreement between the prediction from the model and the present experimental data is good, especially considering the priori nature of this work. Of course, further experimental and theoretical efforts are still needed to develop a kinetic model of high-fidelity.

4.3. Kinetic model validation against experimental data in the literature

The model was used to simulate the datasets in previous studies using the constant volume batch reactor model. To our knowledge, IDT measurements for PMH include the work of Won et al. [10] and Min et al. [8]. Figures 10 and 11 present the comparison between the simulations and experimental data. As shown in Fig. 10, the agreement between the predictions and experimental results from [10] is very good under all conditions, including the NTC region.

The model well predicts the variation of the measured IDTs by Min et al. [8], including the temperature window of NTC behavior. However, discrepancies are found between the experimental data in reference [8] and the predictions, as shown in Fig. 11(a) and (c). Generally, the active radicals produced during the compressions stroke in RCM measurements and the heat loss might have a significant influence on the measured IDTs. Note that the test conditions in the two mentioned panels are either at low pressure (10 bar) or under fuel-lean conditions, such that the IDTs are relatively long and the results tend to be affected by above factors. Due to the unavailability of the pressure history of the non-reactive runs, the simulations were performed using constant volume constraint. Therefore, these discrepancies can be expected and are acceptable.

4.4. Kinetic analyses

To date, knowledge about the combustion process of PMH is poor. In order to elucidate the reaction channels that control the

system, kinetic analyses, including reaction pathway analysis and sensitivity analysis, were carried out for better understanding of the combustion chemistry of PMH using the model proposed in the present study.

4.4.1. Reaction pathway analysis

Reaction pathway analysis was performed for stoichiometric PMH/air mixture at the time corresponding to 20% fuel conversion under 20 bar. Three temperatures of 650 K, 800 K and 1300 K were selected to represent the low, intermediate and high temperature region, respectively. Figure 12 shows the results of the reaction analysis.

As shown in Fig. 12(a), the fuel is mainly consumed by H abstraction reactions at both 650 and 800 K, although a slight variation in the branching ratios producing different fuel derived radicals is observed at the two temperatures. Among the abstractors, the OH radicals contribute the most for the conversion of PMH. H abstraction by HO₂ contributes little to fuel consumption at 650 K, but the importance of this channel increases as temperature rises. This is consistent with the increasing importance of the decomposition of H₂O₂ in the NTC region. PMH-R3 is the major formed fuel derived radical due to the presence of 4 equivalent secondary H atoms at this site. While for PMH-R1, although there are 18 H atoms for abstraction at carbon 1, the strong primary C–H bonds make this channel less favored than the formation of PMH-R3. The preferred reaction channel for fuel derived radicals at 650 and 800 K is the oxygen addition to form alkylperoxy radicals, which accounts for over 79% consumption for all the fuel derived radicals. By contrast, the beta scission pathway can be neglected. Except for PMH-R1O2 which has no H atom to the beta carbon of the O–O group, other alkylperoxy radicals are mainly consumed by two competition reaction channels, namely the isomerization of ROO to QOOH species and the concerted elimination to produce the relatively stable alkenes. Figure 12(a) shows that the contributions of concerted elimination increase as the temperature rises from 650 K to 800 K, implying the reduced overall reactivity due to the formation of more stable olefins. Therefore, the competition of these two channels is generally considered to be one of the reasons for NTC behavior. Note that the concerted reaction for PMH-R4O2 is most favored due to the absence of the 6-membered ring isomerization state and more available H atoms on the beta carbons for this tertiary peroxy radical. Therefore, it can be speculated that the formation of PMH-R4 would inhibit the reactivity. Since the concerted elimination pathway is impossible for PMH-R1O2, the formation of PMH-R1 may accelerate the chain branching process and show stronger promotion effect on the reactivity, compared with

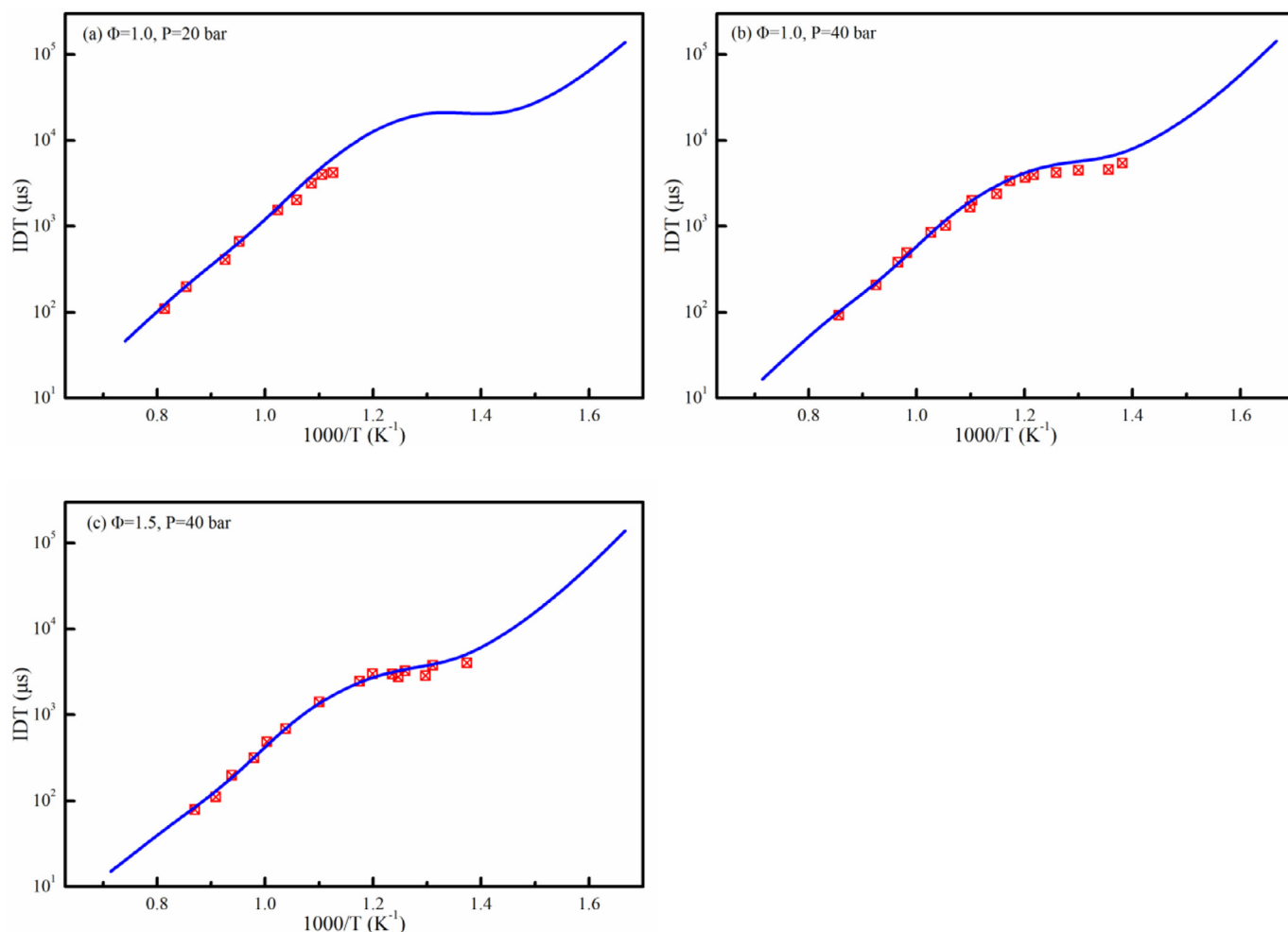


Fig. 10. Comparison of predicted IDTs from the present model with experimental data in [10]. Symbols are experimental data, lines are simulations.

the other three radicals. Since the QOOH species are too numerous to list one by one, only the reaction pathways of three representative QOOH species are presented in Fig. 12(a). The three shown species are all γ -QOOH, they are mainly consumed by three reaction pathways, including the second O_2 addition, the formation of cyclic ether and the decomposition of QOOH. Out of the three channels, the second O_2 addition shows the predominant flux, but the contributions of the other two channels increase as temperature rises. Since the second O_2 addition is an important step of the low temperature chain branching reactions sequence, the more effective competition from the other two channels at the higher temperature would lead to decreased reactivity. The further reactions of the OOQOOH, e.g., the isomerization to ketohydroperoxides and the further decomposition of ketohydroperoxides, finalize the low temperature chain branching process but are not depicted here to keep the size of the graph at a reasonable level.

As temperature rises to 1300 K, the H abstraction reactions still dominate the fuel conversion, as shown in Fig. 12(b). Compared with the low-to-intermediate temperature range, the importance of H abstraction by H radicals is largely increased since the fuel derived radicals are readily to produce H radical via beta scission and the subsequent reactions. The unimolecular decomposition also consumes a significant part of fuel due to the more available energy at high temperature to overcome the required high energy barrier. The dominant consumption pathway for fuel derived radicals is beta scission to produce smaller radicals and alkenes, while the fuel-related reactions contribute less to the system. Consequently, the small species chemistry controls the overall

reactivity and thus mixtures with different equivalence ratios show very close reactivity at high temperature range. Note that the oxygen addition still occurs at 1300 K, but the products are mainly consumed via concerted elimination to form olefins rather than isomerization to QOOH species.

4.4.2. Sensitivity analysis

To elucidate the key reactions responsible for the autoignition of PMH under different conditions, brute force sensitivity analysis was performed at 20 bar for stoichiometric PMH/air mixture. In order to be consistent with the above section, 650, 800 and 1300 K were selected to present the low, intermediate and high temperature range, respectively. Furthermore, as shown in Fig. 5, the model underestimated the IDT for fuel-lean mixtures at 15 bar in intermediate temperature range obviously, therefore, sensitivity analysis was performed for equivalence ratio of 0.5 at 800 K to identify key reactions for improving the performance of the model. In this study, the sensitivity coefficient was defined as: $S_i = [\tau(2k_i) - \tau(k_i)]/\tau(k_i)$. Here, k_i represents the rate constant of reaction i , $\tau(k_i)$ and $\tau(2k_i)$ denote the predictions when the rate constant of reaction i was set as k_i and $2k_i$, respectively. As such, a negative coefficient implies that the corresponding reaction promote the autoignition and vice versa.

Figure 13 shows the reactions with top sensitivity coefficients under different temperatures. At 650 K, the key reactions showing promoting effect mainly comprise of the fuel-related reactions, including the H abstraction from PMH, the oxygen addition, the isomerization of alkylperoxy, the second oxygen addition and the

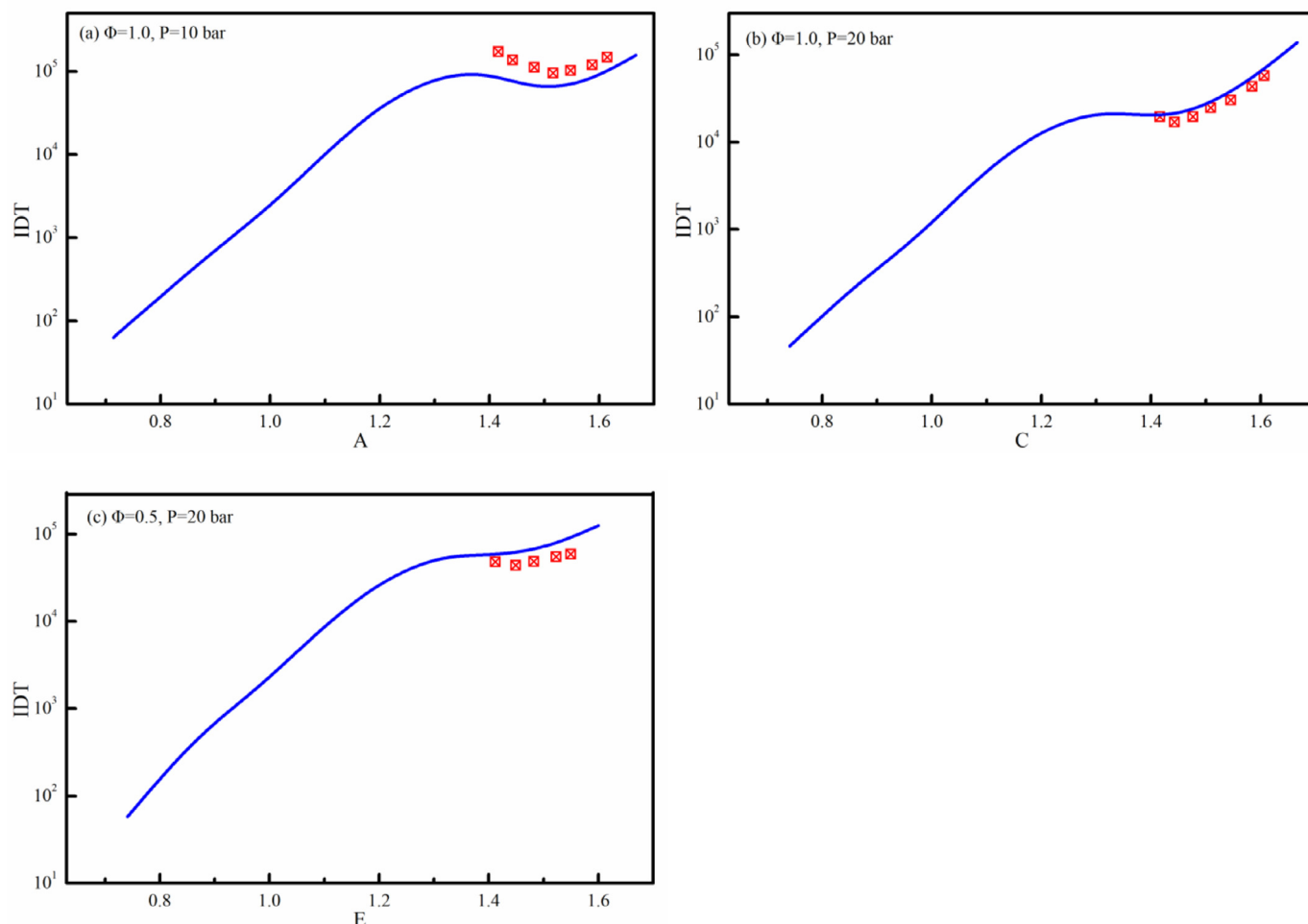
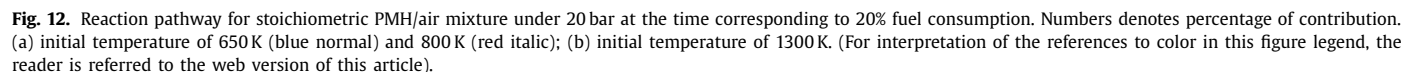


Fig. 11. Comparison of predicted IDTs from the present model with experimental data in [8]. Symbols are experimental data, lines are simulations.

isomerization of OOQOOH to ketohydroperoxides. Note that most of the listed reactions are related to PMH-R3, which is consistent with the largest reaction flux of this fuel derived radical shown in the pathway analysis section. Overall, the isomerization of ROO species shows the highest promoting effect. Among these reactions, the isomerization of PMH-R3O2 to PMHOOH3-6, involving intra-molecular hydrogen transfer of secondary H via 6-membered ring transition states, shows the most promoting effect. By contrast, the formation of PMHOOH3-1 shows much lower sensitivity since H atoms involved in the isomerization process are stronger primary H atoms. On the other hand, the concerted elimination of PMH-R3O2 shows the strongest inhibiting effect on the reactivity through producing less reactive olefins. The formation of PMH-R4 via H abstraction by OH shows inhibiting effect on the autoignition as well. Due to no available H atoms on the tertiary carbon, the related OOQOOH species tends to undergo an alternative isomerization rather than isomerization to ketohydroperoxide via conventional isomerization, rendering this reaction pathway less favored. At 800 K, the fuel related reaction still plays a significant role, although with a variation in the sensitivity coefficients. A notable variation is that the competition between reactions $\text{H}_2\text{O}_2(+\text{M})=\text{OH}+\text{OH} (+\text{M})$ and $\text{HO}_2+\text{HO}_2=\text{H}_2\text{O}_2+\text{O}_2$ become effective and plays a critical role in the determination of the autoignition. This is because the reaction $\text{H}_2\text{O}_2(+\text{M})=\text{OH}+\text{OH} (+\text{M})$ is endothermic and only becomes significant at higher temperatures. In Fig. 13(b), the concerted elimination and the formation of cyclic ether, which are believed to be responsible for the NTC behavior, show high negative sensitivities to the autoignition. In compari-

son, the system is less sensitive to the formation of cyclic ether. Generally, the produced HO_2 and OH radicals in the above two reactions can both promote fuel consumption via H abstraction reactions and compensate the decreased reactivity to some extent. However, the HO_2 radical is much less reactive and contributes little to the fuel conversion at low temperature, and thereby the concerted elimination reactions still show strong inhibiting effect on the system. It is worth noting that the formation of PMH-R1 shows the strongest sensitivity at both 650 and 800 K, compared to the other three fuel derived radicals, which demonstrates the speculation of the pathway analysis section, namely, the formation of PMH-R1 shows stronger promoting effect on autoignition. At 1300 K, the low temperature chain branching pathways become negligible, and the chemistry of small molecular dominates the reactivity of the system. We note that the unimolecular decomposition of fuel ($\text{DC8H17}+\text{TC4H9}=\text{PMH}$) shows stronger promoting effect compared with in the low-to-intermediate temperature range, which is consistent with the increasing importance of beta-scission of radicals as temperature rises, as shown in Fig. 12.

Figure 14 shows the key reactions for the autoignition of PMH/air mixtures with equivalence ratio of 0.5 under 15 bar at 800 K. Overall, the results are very similar to that for the stoichiometric mixture at 15 bar, although subtle difference is observed. The reactions $\text{H}_2\text{O}_2(+\text{M})=\text{OH}+\text{OH} (+\text{M})$ and $\text{HO}_2+\text{HO}_2=\text{H}_2\text{O}_2+\text{O}_2$ shows the highest promoting and inhibiting effects, respectively. The H abstraction from the parent fuel to produce PMH-R1 is found to be sensitive to the systematic reactivity. This may be attributed to two factors: firstly, there are 18 H atoms out of 26 H



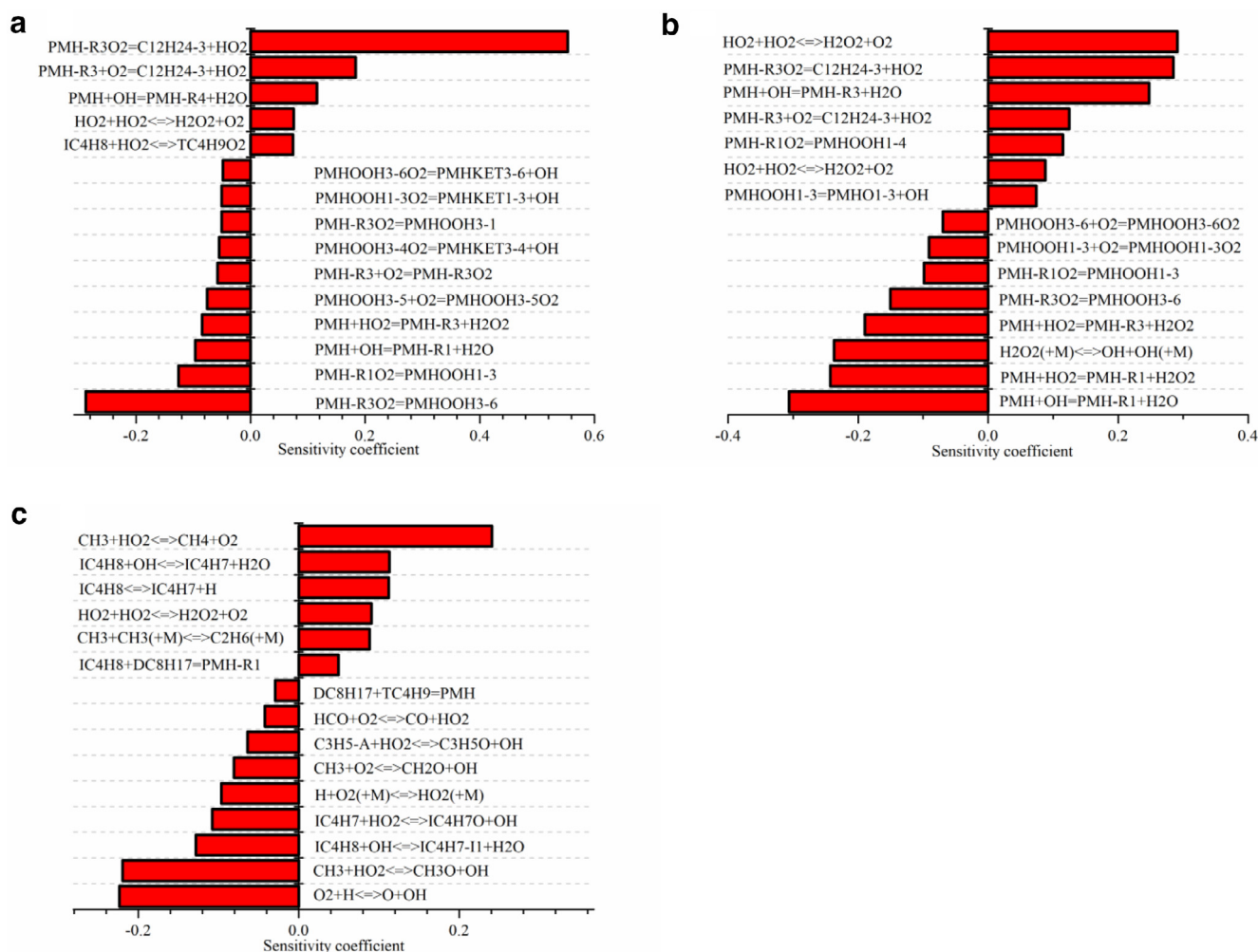


Fig. 13. Sensitivity analysis at 20 bar for stoichiometric PMH/air mixture with different initial temperature. (a) 650 K, (b) 800 K and (c) 1300 K.

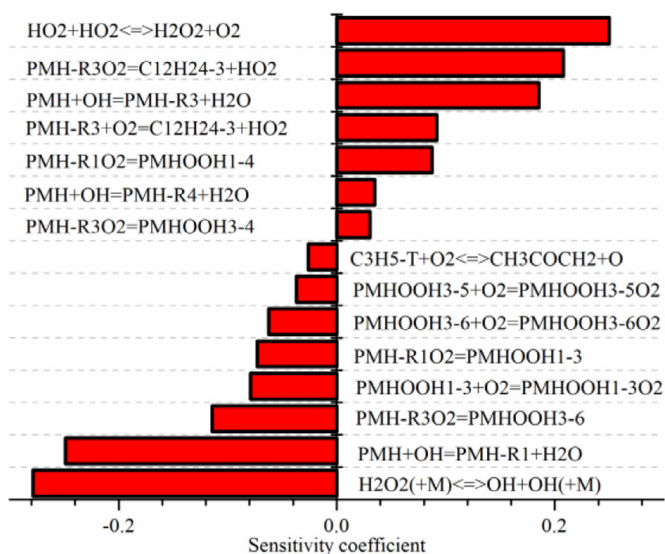


Fig. 14. Sensitivity analysis under 15 bar for equivalence ratio of 0.5 at 800 K.

atoms that can be abstracted to form PMH-R1; and secondly, the related alkylperoxy radical could not undergo the concerted elimination, which makes this channel preferred. Furthermore, the isomerization of PMH-R3O₂ and the subsequent second O₂ addition

shows high sensitivity to the autoignition. The rate constants of these fuel-related reactions are all prescribed based on analogies and thus may have great uncertainties. Therefore, more fundamental kinetics studies on these reactions are essential to improve the performance of the present model.

5. Effect of alternative isomerization and 3rd O₂ addition pathway

The alternative isomerization pathway was found to have minor contributions to the reactivity of linear-alkanes in the study of Zhang et al. [33]. However, the situation would be different for branched alkanes due to the presence of the tertiary carbons. The alternative isomerization and the followed 3rd O₂ addition, as well as the subsequent reactions, were included in the present model, leading to the additional 88 species and 140 reactions. To evaluate the effect of the inclusion of this reaction pathway, a comparison between predictions using models with and without the alternative isomerization channel is presented in Fig. 15. The results show that the inclusion of this reaction pathway leads to a slight reduction in reactivity at low temperature but a significant increase at intermediate temperatures. This is in accord with the findings of Bugle et al. and Zhang et al. [43,45]. Traditionally, only migration of the weakest hydrogen at the α -site to the peroxy group was taken into account for isomerization of peroxy-hydroperoxy alkyls (OOQOOH). With the inclusion of alternative isomerization pathway, stronger H atoms at other sites can also be

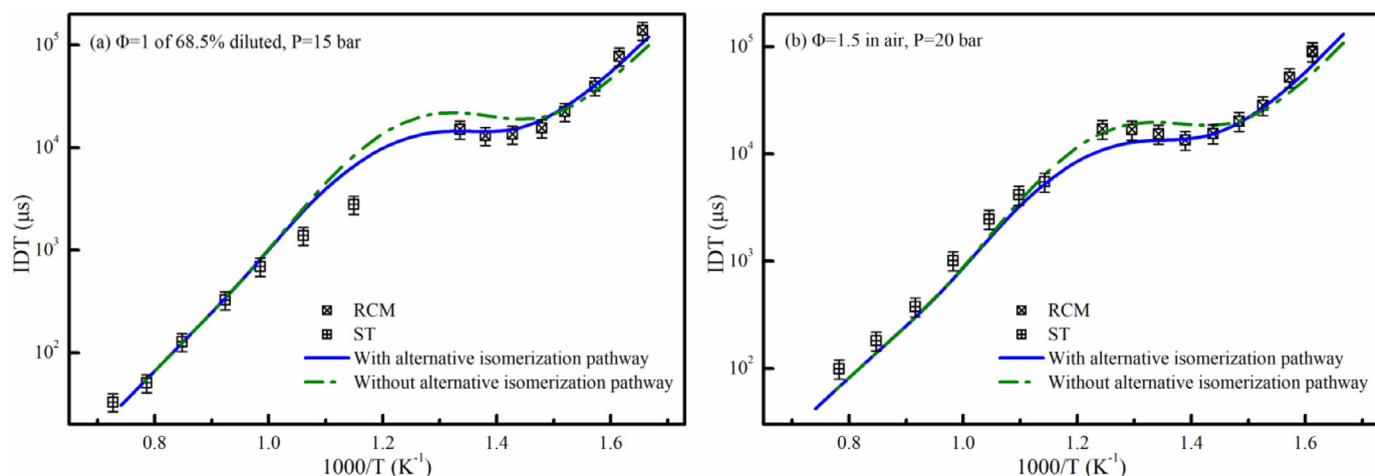


Fig. 15. Comparison of the performance of the models with and without alternative isomerization pathway.

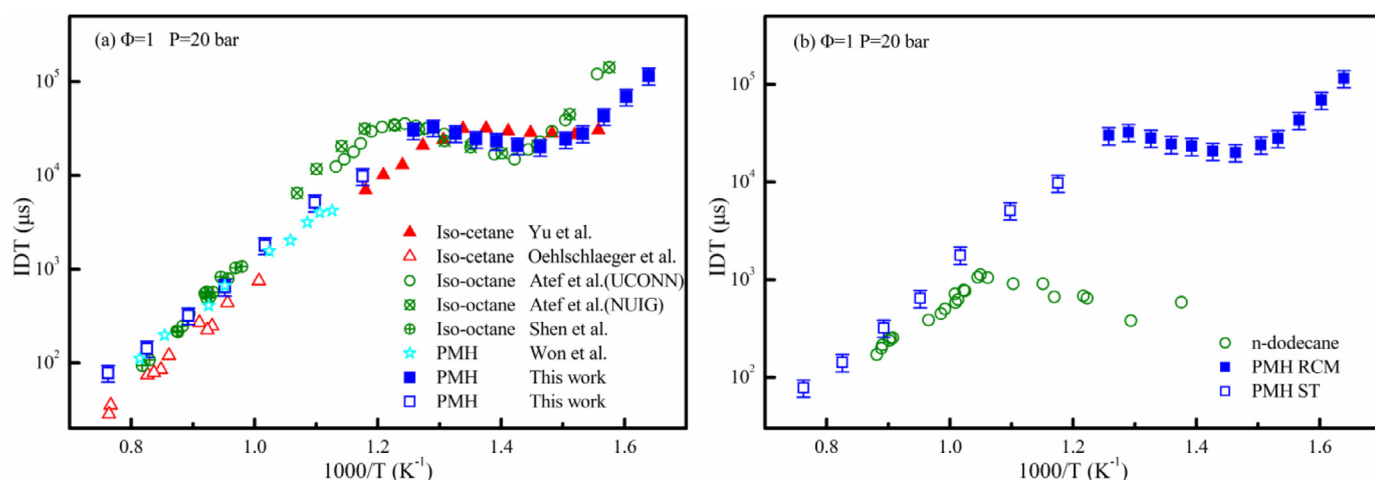


Fig. 16. Comparison of measured IDTs for PMH with for analogues in literature [24,35,37,48,49].

abstracted. Therefore, this newly added channel is slower than the conventional isomerization pathway and competes with it, leading to a reduced reactivity. Another factor leading to the reduced reactivity at low temperatures is the further decomposition of the formed $P(OOH)_2$ species. Compared with 2 OH radicals produced from the conventional isomerization pathway, the decomposition of $P(OOH)_2$ and the successive decomposition of the produced olefinic hydroperoxide produce a HO_2 and OH. Given that OH radical is more reactive than HO_2 radical, the overall reactivity decrease. However, at intermediate temperatures, HO_2 is converted to H_2O_2 after H abstraction and soon decomposes into two OH radicals, and thus promotes the systematic reactivity. Furthermore, the inclusion of this new channel renders further chain propagation possible for the $OOQOOH$ species stemming from a tertiary alkyl radical. Consequently, the reactivity increases at the intermediate temperatures. Overall, as shown in Fig. 15, the addition of the alternative isomerization and the subsequent reactions lead to an improved performance of the model.

6. Comparison with other hydrocarbons

The current data was compared with the previous data for hydrocarbon with similar structure or the same molecular formula to reveal the effect of molecular structure on autoignition. Figure 16 shows a comparison of IDTs among PMH, iso-octane, iso-cetane and n-dodecane. Although there is a noticeable similarity in the structures and combustion chemistry among the three

branched alkanes, the results shown in Fig. 16(a) imply that they do show different autoignition characteristics. At high temperature region, the IDTs prolong as the carbon number increases. The difference in reactivity among the three hydrocarbons can be attributed to several factors. The first factor is the fraction of the primary H atoms accounted for the total H atoms in the parent fuel. Throughout the entire temperature range, the fuels are mainly consumed by H atom abstraction. Therefore, the rate of this reaction would affect the systematic reactivity to a large extent. It is well known that the primary H atoms have the strongest C–H bond and are most difficult to be abstracted. It can be seen from Fig. 3 that the fractions of primary H atoms are 15/18 for iso-octane, 21/26 for PMH and 27/34 for iso-cetane, respectively. The lower fraction of primary H atoms implies that the fuel is more reactive under the same conditions. Secondly, the thermal stability of the parent fuels also affects the IDTs, since the unimolecular decomposition plays an increasingly important role as temperature increases. As the carbon chain increases, the thermal stability decreases for these fuels. Thirdly, the recombination of CH_3 , a chain termination reaction, contributes to the discrepancy in reactivity. The proportions of CH_3 in the three fuels increase in the order of iso-cetane, PMH and iso-octane. The combination of the above-mentioned factors leads to the increasing reactivity as carbon number increases in the high temperature range. At very high temperatures, the difference of IDTs shows no clear trend partly due to the very short IDTs which tend to be interfered readily by the non-ideal environments. In the intermediate temperature

region, the reactivity of the three fuels is in the opposite order, compared with the high temperature range. The reason is yet not clear and it could be a subject of interest in future work. An interesting observation is that the three alkanes show distinct NTC region, i.e., the NTC temperature window shifts to higher temperature as carbon number decreases. Figure 16(b) present a comparison between the IDTs of PMH and of n-dodecane. It is found that n-dodecane shows very different combustion chemistry from PMH, especially in low-to-intermediate temperature regime. Given the disparate structures, the difference in Fig. 16(b) can be expected, because the structure of n-dodecane allows more facile abstraction of “secondary” H-atoms from the gamma carbon during the isomerization of ROO compared to PMH.

7. Conclusion

In this study, the autoignition of PMH, an important surrogate component candidate for jet fuels and the main component in alternative fuels, has been systematically studied over low-to-high temperature range at elevated pressures using a heated RCM and ST. Typical double-stage ignition and NTC behavior were observed for all the tested mixtures at different pressures. The effects of pressure, fuel molar fraction and oxygen molar fraction were systematically presented. The results demonstrate that the increase of the above-mentioned parameters tends to decrease the IDTs of PMH and show the most pronounced effect in NTC region. The effect of dilution ratio was also investigated by varying nitrogen loading at a fixed equivalence ratio, i.e., increasing dilution ratio would decrease the overall reactivity. Furthermore, a detailed kinetic model was developed based on the rate rules employed in previous literature to describe the oxidation of PMH, including the high temperature and low temperature reaction schemes. Overall, the agreement between the predictions from the model and the experimental data is satisfactorily good, especially in light of the complex structure of the target fuel. However, discrepancies were still observed between the simulations and experimental data, since this priori model was developed mainly based on the rate rules and analogies. Therefore, further experimental and theoretic studies, especially on the more fundamentally rate coefficients and thermodynamic data, are needed for the development of a detailed mechanism of more high fidelity. Kinetic analyses were performed to identify the key reaction pathway during the oxidation of PMH. It is found that the formation of radical PMH-R3 is the most favored channel for fuel consumption. In low-to-intermediate temperature range, the competition between isomerization of ROO and concerted elimination controls the reactivity, while at high temperatures, the small species chemistry, especially reaction $H + O_2 = OH + O$ dominates the autoignition. The experimental data and kinetic model would contribute to a good understanding of the combustion chemistry of this highly branched alkane and would be useful for developing surrogates for jets fuels and alternative fuels.

Acknowledgments

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2019.08.030.

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